

CPTG Native-Phase Structural Test

Effective-Radius Closure and Structural Mode N of ZF-UDS-7329

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Abstract

ZF-UDS-7329 (PRIMER-EXCELS-55410) is a massive, compact quiescent galaxy for which the EXCELS source paper reports $z = 3.1954 \pm 0.0002$, $\log_{10}(M_{\star}/M_{\odot}) = 11.19 \pm 0.02$, and a central formation marker $z_{\text{form}} = 14.5^{+8.9}_{-4.2}$. This paper separates the early-galaxy problem into cosmological phase, bound-system field support, and radial structural organization. The formation marker maps to CPTG native phase $N_{\pi, \text{form}} = -2.74084$ and native clock coordinate $t_{\pi, \text{form}} = 0.21451$ Gyr; from formation to observation, native phase accumulates about 6.97 times faster per gigayear than from observation to the normalized present coordinate. The structural calculation uses the published EXCELS F277W size/profile, intrinsic stellar dispersion, and effective-radius dynamical measurements for the same galaxy. At the observed phase, the CPTG-coordinate effective radius is $R_e = 0.90073$ kpc and the declared pressure-support estimator gives $a_{\star} \simeq (2.249 \pm 0.361) \times 10^{-9} \text{ m s}^{-2}$. The stellar source gives $g_{\text{CPTG}, e} = (1.498 \pm 0.074) \times 10^{-8} \text{ m s}^{-2}$, closing against the correspondingly converted published virial comparator at a nominal shared-input residual of $+0.06\sigma$. Structural organization is measured by the weighted centroid λ of radial field change and $N = R/\lambda$. The spherical reconstruction gives $\lambda = 1.45404$ kpc and $N = 3.33030$; inserting the measured flattening gives $\lambda = 1.39593$ kpc and $N = 3.46894$. CPTG thereby resolves the gravitational-accounting and native-timing components of the early-galaxy problem while leaving star-formation efficiency and quenching microphysics as separate astrophysical questions.

1 Introduction: Why ZF-UDS-7329 Matters

ZF-UDS-7329 is a compact quiescent galaxy at high redshift whose early maturity makes it an unusually stringent galaxy-formation target. The EXCELS source paper reports $z = 3.1954 \pm 0.0002$ and $\log_{10}(M_{\star}/M_{\odot}) = 11.19 \pm 0.02$, with a central formation marker $z_{\text{form}} = 14.5^{+8.9}_{-4.2}$ and quenching marker $z_{\text{quench}} = 9.90^{+7.79}_{-3.29}$ [1]. The galaxy is therefore already massive and quiescent at an exceptionally early stage of the observed history.

The difficulty is not one observation in isolation but the sequence that must occur quickly: a very large stellar mass must assemble, the stellar system must become compact, and star formation must then shut down. The source paper also gives a 10-Myr-averaged SFR upper limit of $1.13 M_{\odot} \text{ yr}^{-1}$, $\log_{10}(Z_{\star}/Z_{\odot}) = 0.07^{+0.14}_{-0.09}$, $A_V = 0.26 \pm 0.06$, a 10–90 percent assembly width $\tau_{10-90} = 262^{+287}_{-200}$ Myr, and $[\text{Mg}/\text{Fe}] = 0.24 \pm 0.10$ [1]. These are

population and chemical constraints on a galaxy that is already dynamically measurable at the observed phase.

The same object is unusually useful for CPTG because its redshift selects a well-defined native geometric phase, while the published EXCELS F277W size, Sérsic profile, ellipticity, intrinsic stellar dispersion, and effective-radius dynamical estimate provide the structural quantities needed for a bound-system test [2]. CPTG therefore asks a different sequence of questions from a halo-assembly model: where does the native π branch place the galaxy; what object-dependent structural acceleration scale $a_\star$ is measured by the galaxy; does the same bound-system field law close at the observed effective radius; and where is the solved radial response organized through the Structural Mode organization length λ and the dimensionless mode N ?

The distinction is important. CPTG does not make an early galaxy possible by rescaling its intrinsic $a_\star$ with the Hubble rate, and it does not require a dark-matter halo to supply the galaxy-scale field. The redshift sets the cosmological phase first; the observed galaxy then supplies its own structural measurement. Section 4 makes explicit how that changes the gravitational and timing interpretation of the early-galaxy problem, while also stating what CPTG does *not* claim to solve.

2 Purpose, Test Definition, and Observed Inputs

The calculation separates cosmological placement from bound-system structural measurement. ZF-UDS-7329 is not assigned a special high-redshift gravity law, and its intrinsic structural acceleration is not rescaled merely because the galaxy is observed at large redshift. The operational sequence is [7, 8]

$$\begin{array}{c}
 \underbrace{z \longrightarrow N_\pi \longrightarrow \text{selected epoch and branch coordinates}}_{\text{cosmological placement}} \\
 \Downarrow \\
 \underbrace{\text{galaxy structure} \longrightarrow a_\star \longrightarrow \{\text{field closure, } N, \text{ CSMI}\}}_{\text{structural measurement}}
 \end{array} \tag{1}$$

The first line answers *when* the galaxy is represented in the CPTG geometry. The second answers *what the galaxy itself measures* at that selected phase. In particular, this paper does not impose a universal $a_\star \propto H$ law and does not divide the observed support coordinate by an expansion function to manufacture a present-equivalent $a_\star$.

The effective-radius comparison is deliberately a closure/consistency test rather than an independent prediction test. Carnall et al.'s published virial mass is constructed from the same compact size and stellar dispersion that participate in the CPTG support estimate, so the two sides necessarily share input information [2].

2.1 Observed inputs and distance convention

The EXCELS source paper supplies the adopted population parameters: $z = 3.1954 \pm 0.0002$, $\log_{10}(M_*/M_\odot) = 11.19 \pm 0.02$, $\text{SFR}_{10\text{ Myr}} < 1.13 M_\odot \text{ yr}^{-1}$, $\log_{10}(Z_*/Z_\odot) = 0.07^{+0.14}_{-0.09}$, $t_{\text{form}} = 0.28^{+0.17}_{-0.14}$ Gyr in the source paper’s conventional age coordinate, $z_{\text{form}} = 14.5^{+8.9}_{-4.2}$, $z_{\text{quench}} = 9.90^{+7.79}_{-3.29}$, $A_V = 0.26 \pm 0.06$, $\tau_{10-90} = 262^{+287}_{-200}$ Myr, and $[\text{Mg}/\text{Fe}] = 0.24 \pm 0.10$ [1]. These values are the observational authority used for the population and phase quantities throughout this paper.

The source paper’s spectral-broadening fit parameter is not used as the intrinsic stellar velocity dispersion because the model and data spectral resolutions are not separated in that quantity [1]. For the bound-system calculation, the adopted EXCELS structural measurements for ZF-UDS-7329 are $\sigma_* = 250 \pm 20 \text{ km s}^{-1}$, F277W $R_e = 0.910 \pm 0.010$ kpc, Sérsic index $n = 2.5 \pm 0.1$, ellipticity $e = 0.77 \pm 0.01$, and $\log_{10}(M_{\text{dyn},e}/M_\odot) = 10.94 \pm 0.07$ [2]. The F277W profile was fitted with PetroFit using the Astropy two-dimensional Sérsic model [3, 4]; in that parameterization R_e is the semimajor-axis half-light radius and $e = 1 - b/a$.

The EXCELS source convention is flat- Λ CDM with $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$ [1]. Those quantities identify the source distance convention only; they do not govern the CPTG branch mathematics.

For the neutral source-unit conversion, the present acoustic and luminosity CPTG comparison endpoints are

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351, \quad H_{0,\text{lum}}^{\text{CPTG}} = 73.17 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (2)$$

whose arithmetic midpoint is

$$\frac{H_{0,\text{ac}}^{\text{CPTG}} + H_{0,\text{lum}}^{\text{CPTG}}}{2} = 70.3238983676 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (3)$$

The EXCELS source value 70 therefore lies close to the center of the two observational CPTG projections. For a neutral source-unit conversion, this paper uses the arithmetic midpoint of the *distance coordinates* themselves, not a newly defined cosmological branch.

For $b \in \{\text{ac}, \text{lum}\}$,

$$D_{M,b}(z) = \frac{c}{H_{0,b}} \int_0^z \frac{dz'}{E_b(z')}, \quad D_{A,b}(z) = \frac{D_{M,b}(z)}{1+z}, \quad (4)$$

and

$$D_{A,\text{mid}}(z) \equiv \frac{D_{A,\text{ac}}(z) + D_{A,\text{lum}}(z)}{2}. \quad (5)$$

At $z = 3.1954$,

$$D_A^{\Lambda 70} = 1558.1095 \text{ Mpc}, \quad (6)$$

$$D_{A,\text{ac}} = 1602.3790 \text{ Mpc}, \quad (7)$$

$$D_{A,\text{lum}} = 1482.4319 \text{ Mpc}, \quad (8)$$

$$D_{A,\text{mid}} = 1542.4055 \text{ Mpc}. \quad (9)$$

Thus the population-fit quantities use

$$s_{D,\text{pop}} \equiv \frac{D_{A,\text{mid}}(3.1954)}{D_A^{\Lambda 70}(3.1954)} = 0.98992112. \quad (10)$$

The published EXCELS structural quantities use the physical scale associated with $z = 3.1943$. To express those measured observables at the source-paper phase $z = 3.1954$ on the CPTG comparison coordinate, their physical scale uses the direct source-to-CPTG ratio

$$s_{D,\text{struct}} \equiv \frac{D_{A,\text{mid}}(3.1954)}{D_A^{\Lambda 70}(3.1943)} = 0.98981082. \quad (11)$$

The effective radius therefore maps as

$$0.910 \pm 0.010 \text{ kpc} \longrightarrow R_e = 0.90073 \pm 0.00990 \text{ kpc}. \quad (12)$$

The change remains only about 1.02%.

At fixed observed flux, the leading distance dependence of the stellar-mass estimate is $M_\star \propto D_L^2$. Because the same fixed redshift multiplies both D_L and D_A , Eq. (10) also gives the luminosity-distance ratio. The total stellar source therefore maps as

$$\log_{10}(M_\star/M_\odot) = 11.19 \pm 0.02 \longrightarrow 11.1812 \pm 0.02. \quad (13)$$

The source paper does not tabulate an aperture stellar mass. Under the same constant stellar mass-to-light assumption used below for the Sérsic radial source, the semimajor-axis half-light radius contains one half of the model stellar mass. The one-radius stellar source is therefore derived consistently from the adopted total mass,

$$\log_{10}(M_{\star,e}/M_\odot) = 11.19 - \log_{10} 2 + 2 \log_{10} s_{D,\text{pop}} = 10.8802 \pm 0.02. \quad (14)$$

This is an aperture-source construction under the declared constant- M/L reduction, not a separately fitted aperture mass.

For the published virial dynamical estimator, fixed $\sigma_\star$ and structural coefficient imply $M_{\text{dyn},e} \propto R_e$. Its direct structural-coordinate conversion is

$$\log_{10}(M_{\text{dyn},e}/M_\odot) = 10.94 \pm 0.07 \longrightarrow 10.9356 \pm 0.07. \quad (15)$$

Quantity	Adopted CPTG-coordinate value	Role in test
Formal target	ZF-UDS-7329	Object identity
Redshift	3.1954 ± 0.0002	Common cosmological phase
Intrinsic stellar dispersion	$250 \pm 20 \text{ km s}^{-1}$	Effective-radius support measurement
F277W effective radius	$0.90073 \pm 0.00990 \text{ kpc}$	Semimajor-axis half-light scale on the CPTG comparison coordinate
Stellar mass within R_e	$\log_{10}(M_{\star,e}/M_{\odot}) = 10.8802 \pm 0.02$	Constant- M/L one-radius source derived from total mass
Total stellar mass	$\log_{10}(M_{\star}/M_{\odot}) = 11.1812 \pm 0.02$	Sérsic radial-source normalization for Structural Mode
F277W Sérsic index	$n = 2.5 \pm 0.1$	Measured radial profile shape
F277W ellipticity	$e = 0.77 \pm 0.01$ ($q_{\text{proj}} = 0.23 \pm 0.01$)	Projected source geometry for the oblate stress test
Published virial dynamical mass	$\log_{10}(M_{\text{dyn},e}/M_{\odot}) = 10.9356 \pm 0.07$	Shared-input closure comparator

Table 1: Inputs used in the ZF-UDS-7329 calculation. The EXCELS source paper supplies the population and phase values; the published EXCELS imaging and kinematic measurements supply the structural observables. Distance-dependent quantities are re-expressed on the symmetric CPTG acoustic/luminosity midpoint distance coordinate.

The observational inputs therefore enter according to their measured roles: the source-paper redshift and stellar-population quantities set the phase and stellar normalization, while the EXCELS imaging and kinematic observables set the bound-system structure.

3 Native Phase and Branch Coordinates

This section answers the first question: *where on the CPTG history is ZF-UDS-7329 observed?* Redshift selects one common native π -branch phase. The acoustic and luminosity quantities below are projections of that same native history, so their different integrated clock coordinates do not describe different galaxies or different universes. For the measured redshift, the common geometric phase is

$$a = \frac{1}{1+z} = 0.2383562950, \quad N_{\pi} = \ln a = -1.433988687. \quad (16)$$

The native π branch and its acoustic and luminosity projections are [7, 8]

$$E_{\pi}(a)^2 = \mathcal{B}_{\pi} + \mathcal{T}_{\pi} a^{-\pi}, \quad \mathcal{B}_{\pi} = 1 - \frac{1}{\pi}, \quad \mathcal{T}_{\pi} = \frac{1}{\pi}, \quad (17)$$

$$E_{\text{ac}}(a)^2 = \mathcal{B}_{\pi} + \mathcal{T}_{\pi} a^{-p_{\text{ac}}}, \quad p_{\text{ac}} = 3 - \frac{\pi}{100}, \quad (18)$$

$$E_{\text{lum}}(a)^2 = \Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}, \quad \Omega_{m,\text{ac}} = \frac{p_{\text{ac}}}{3\pi}, \quad \Omega_{\Lambda,\text{ac}} = 1 - \Omega_{m,\text{ac}}. \quad (19)$$

The corresponding present branch and projection normalizations are

$$\begin{aligned} H_0^{(\pi)} &= 69.4162507897, \\ H_{0,\text{ac}}^{\text{CPTG}} &= 67.4777967351, \quad \text{km s}^{-1} \text{ Mpc}^{-1}, \\ H_{0,\text{lum}}^{\text{CPTG}} &= 73.17 \end{aligned} \quad (20)$$

The first is the native CPTG present-rate normalization; the acoustic value is a derived comparison coordinate; the luminosity value is the data-conditioned luminosity endpoint. They are not three independent cosmological fits.

At the ZF-UDS-7329 phase,

$$E_\pi = 5.429447, \quad E_{ac} = 4.811632, \quad E_{lum} = 4.787466. \quad (21)$$

For any branch of the form $E^2 = B + Ta^{-q}$, the integrated branch-clock coordinate is

$$t_b(a) = \frac{2}{qH_{0,b}\sqrt{B}} \sinh^{-1} \left(\sqrt{\frac{B}{T}} a^{q/2} \right). \quad (22)$$

This gives Table 2. The three rows are different clock coordinates assigned to the same N_π ; they are comparison readouts of one geometric phase, not different physical kinds of time.

Clock coordinate	$t(z = 3.1954)$ (Gyr)	t_0 (Gyr)
Native π branch	1.664530	12.754121
Acoustic projection	2.049235	13.885174
Luminosity comparison	1.899644	12.843057

Table 2: Branch-specific clock coordinates at the common ZF-UDS-7329 geometric phase and at the normalized present phase.

With the observation located cosmologically, the calculation can now turn to the galaxy itself.

4 The “Impossible Early Galaxy” Problem and the CPTG Resolution

The early-galaxy problem is often summarized as “not enough time,” but that phrase combines several distinct assumptions. ZF-UDS-7329 is especially useful because its stellar-population history and bound-system structure are both observationally constrained. CPTG uses the redshift-selected stages on its native geometry rather than importing the source paper’s flat- Λ CDM age coordinate as the CPTG clock.

4.1 The formation challenge

The EXCELS source paper gives the central population reconstruction

$$z_{\text{form}} = 14.5^{+8.9}_{-4.2}, \quad z_{\text{quench}} = 9.90^{+7.79}_{-3.29}, \quad (23)$$

with source-model half-mass formation time

$$t_{\text{form}}^{\text{source}} = 0.28^{+0.17}_{-0.14} \text{ Gyr} \quad (24)$$

and 10–90 percent stellar-assembly width

$$\tau_{10-90} = 262^{+287}_{-200} \text{ Myr} \quad (25)$$

under the source paper’s conventional cosmological age coordinate [1]. These inferred stages remain model-dependent population markers rather than directly observed birth and shutdown instants. The robust physical point is the sequence itself: a stellar mass of $\log_{10}(M_*/M_\odot) = 11.19 \pm 0.02$ is already present and the 10-Myr-averaged SFR is constrained below $1.13 M_\odot \text{ yr}^{-1}$ by the observed phase [1].

In a conventional halo-based interpretation, the same sequence also requires sufficiently rapid halo assembly and a very high conversion of available baryons into stars. CPTG does not inherit that halo prerequisite. These source measurements make the galaxy a stringent timing and gravitational-accounting stress test.

4.2 CPTG uses native phase, not the source paper’s conventional age

For CPTG, the formation and quenching redshifts are used as phase markers without importing the source paper’s conventional cosmic ages. The central formation marker gives

$$a_{\text{form}} = \frac{1}{1 + 14.5} = 0.0645161, \quad N_{\pi, \text{form}} = \ln a_{\text{form}} = -2.74084, \quad (26)$$

and the native π -branch clock from Eq. (22) gives

$$t_{\pi, \text{form}} = 0.21451 \text{ Gyr}. \quad (27)$$

The quenching marker lies at

$$N_{\pi, \text{quench}} = -2.38876, \quad t_{\pi, \text{quench}} = 0.37288 \text{ Gyr}, \quad (28)$$

while the observed galaxy lies at

$$N_{\pi, \text{obs}} = -1.43399, \quad t_{\pi, \text{obs}} = 1.66453 \text{ Gyr}. \quad (29)$$

The relevant quantity is the cadence of native geometric phase,

$$\frac{dN_\pi}{dt_\pi} = H_\pi(a) = H_0^{(\pi)} E_\pi(a), \quad \Delta N_\pi = \ln\left(\frac{a_2}{a_1}\right). \quad (30)$$

At the central formation marker,

$$\frac{H_\pi(z_{\text{form}})}{H_0^{(\pi)}} = 41.81, \quad (31)$$

at the central quenching marker it is

$$\frac{H_\pi(z_{\text{quench}})}{H_0^{(\pi)}} = 24.06, \quad (32)$$

and at the observed phase it remains

$$\frac{H_\pi(z_{\text{obs}})}{H_0^{(\pi)}} = 5.429. \quad (33)$$

From the central formation marker to observation,

$$\Delta t_\pi = 1.45002 \text{ Gyr}, \quad \Delta N_\pi = 1.30685, \quad \frac{\Delta N_\pi}{\Delta t_\pi} = 0.90126 \text{ Gyr}^{-1}. \quad (34)$$

From the observed phase to the normalized present native coordinate,

$$\Delta t_\pi = 11.08959 \text{ Gyr}, \quad \Delta N_\pi = 1.43399, \quad \frac{\Delta N_\pi}{\Delta t_\pi} = 0.12931 \text{ Gyr}^{-1}. \quad (35)$$

The formation-to-observation interval therefore accumulates native phase about

$$\frac{0.90126}{0.12931} = 6.97 \quad (36)$$

times faster per gigayear than the observation-to-present interval. Even the lower 1σ formation-redshift marker, $z_{\text{form}} = 10.3$, leaves the corresponding cadence ratio at about 5.84. The faster early native cadence is therefore not a consequence of using only the central high- z_{form} value.

This is the CPTG resolution of the geometric timing part of the “not enough time” objection. CPTG does not manufacture additional ordinary years. It states that a gigayear is not a uniform measure of geometric evolution across the native history: substantially more native phase is traversed per gigayear in the early universe than near the normalized present. The inferred formation phase lies deep in precisely that high-cadence native regime.

4.3 CPTG removes the dark-halo prerequisite

The second change is gravitational rather than chronological. In standard galaxy formation, the inferred stellar mass is commonly translated into a required host dark-matter halo, and the rarity or assembly time of that halo becomes part of the early-galaxy problem. CPTG does not contain that prerequisite. The galaxy-scale field is sourced by the observed baryonic structure together with the curvature-polarization/transport response. The object-dependent structural scale $a_\star$ is measured from the galaxy state itself; it is not supplied by a dark halo and is not set by $H(z)$.

For ZF-UDS-7329 this statement is testable at the observed phase. The same bound-system law used for lower-redshift galaxies accepts the measured compact source and pressure support and gives the effective-radius closure developed in the next section. No special early-time gravity law and no dark-matter halo term are added. The existence of a massive, dynamically mature galaxy at $z = 3.1954$ is therefore not a contradiction in CPTG merely because a conventional halo of the corresponding inferred mass would be rare at its formation epoch.

4.4 What CPTG does and does not resolve

CPTG resolves the *gravitational accounting* and the *native geometric timing* parts of the early-galaxy problem. It provides a faster early native-phase cadence, does not require a

dark halo to assemble before the galaxy can possess its measured field, and applies the same object-dependent structural law at the observed high-redshift phase that is used for bound systems more generally.

It does not replace the baryonic microphysics of star formation, stellar feedback, gas cooling, chemical enrichment, or quenching. The EXCELS constraints make those questions more sharply specified but do not turn them into gravitational degrees of freedom. The narrower result established here is that ZF-UDS-7329 need not be treated as “too early to be gravitationally mature” inside CPTG: its cosmological phase and its bound-system structure are separate quantities, and both can be represented self-consistently without a special high-redshift gravity law.

5 Effective-Radius Structural Closure

The cosmological placement does not determine $a_\star$. The local structural measurement comes from the observed galaxy. This section first constructs the one-radius pressure-support estimator and then asks whether the resulting CPTG field is consistent with the published virial comparator.

5.1 Pressure-support measurement

The measured stellar dispersion and converted effective radius define the observable pressure-support coordinate

$$a_\sigma(R_e) = \frac{\sigma_\star^2}{R_e}. \quad (37)$$

The CPTG core framework does not identify a_σ and $a_\star$ by definition. In this declared one-radius pressure-supported reduction, Eq. (37) is adopted as the estimator of the object-dependent structural acceleration scale $a_\star$.

Using the midpoint-converted radius gives

$$a_\star \simeq a_\sigma = \frac{(250 \text{ km s}^{-1})^2}{0.90073 \text{ kpc}} = (2.2487 \pm 0.3606) \times 10^{-9} \text{ m s}^{-2}. \quad (38)$$

At fixed central radius, the published dispersion bracket gives

$$a_\sigma(230 \text{ km s}^{-1}) = 1.9033 \times 10^{-9} \text{ m s}^{-2}, \quad (39)$$

$$a_\sigma(270 \text{ km s}^{-1}) = 2.6229 \times 10^{-9} \text{ m s}^{-2}. \quad (40)$$

The corresponding *local* CPTG curvature-transport radius is

$$R_c = \frac{48\pi c^2}{a_\star} = 195.32 \text{ Gpc}. \quad (41)$$

Here R_e is the observed galaxy effective radius and lowercase R_c is the local bound-system CPTG curvature scale. Uppercase R_C is reserved for the universe-scale native cosmological curvature-transport radius and is not used in this local galaxy reduction.

The number to carry into the reduced field solve is therefore the measured-epoch structural scale in Eq. (38); the branch clocks above do not modify it.

5.2 Field solve and published comparator

The next question is direct: *does the CPTG field generated from the observed support and stellar source remain consistent with the published effective-radius dynamical estimate?* Within the observed effective-radius aperture, the reduced calculation uses the stellar-dominated baryonic source approximation

$$g_{\star,e} = \frac{GM_{\star,e}}{R_e^2}. \quad (42)$$

This is not a full deprojected baryonic mass model. It is the reduced source approximation appropriate to the present one-radius closure, motivated by the compact stellar-dominated central region reported by EXCELS [2].

After the common distance conversion,

$$g_{\star,e} = (1.3038 \pm 0.0666) \times 10^{-8} \text{ m s}^{-2}. \quad (43)$$

The near-invariance of this value under the distance conversion follows from the leading scalings $M_{\star,e} \propto D_L^2$ and $R_e^2 \propto D_A^2$ at fixed redshift.

The reduced CPTG response is solved from

$$g = g_{\star} + \frac{\sqrt{a_{\star}g_{\star}} + g_{\text{geom}}}{\left[1 + (g/a_{\star})^{123/50}\right]^{11/50}}, \quad (44)$$

with

$$g_{\text{geom}}(r) = \frac{1}{2}a_{\star} \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad R_c = \frac{48\pi c^2}{a_{\star}}. \quad (45)$$

At $r = R_e$, Eqs. (38)–(44) give

$$g_{\text{CPTG},e} = (1.4975 \pm 0.0741) \times 10^{-8} \text{ m s}^{-2}. \quad (46)$$

At this compact radius the explicit finite- R_c term is numerically tiny,

$$g_{\text{geom}}(R_e) = 2.40 \times 10^{-21} \text{ m s}^{-2}, \quad (47)$$

so the present one-radius closure is controlled by the baryonic source and the $\sqrt{a_{\star}g_{\star}}$ curvature-polarization/transport response rather than by the finite- R_c contribution itself.

The quoted field uncertainty is propagated through the full implicit solve by applying symmetric one-at-a-time $\pm 1\sigma$ perturbations to $\log_{10} M_{\star,e}$, $\sigma_{\star}$, and R_e , taking the half-range

of each solved-field response, and combining those three input contributions in quadrature. This propagation convention is used only to describe the local measurement sensitivity; it does not remove the shared-input covariance with the published virial comparator.

The correspondingly converted published virial dynamical comparator is

$$g_{\text{dyn},e} = \frac{GM_{\text{dyn},e}}{R_e^2} = (1.4811 \pm 0.2420) \times 10^{-8} \text{ m s}^{-2}. \quad (48)$$

Hence

$$g_{\text{CPTG},e} - g_{\text{dyn},e} = +1.634 \times 10^{-10} \text{ m s}^{-2}. \quad (49)$$

If the quoted uncertainties are combined as though independent, the nominal residual is

$$\Delta_{g,e}^{\text{nominal}} = +0.06\sigma. \quad (50)$$

In mass space, the CPTG field corresponds to

$$\log_{10}(M_{\text{CPTG},e}/M_{\odot}) = 10.9403 \pm 0.0195, \quad (51)$$

where the mass-space uncertainty is propagated from the same raw input perturbations through $M_{\text{CPTG},e} = g_{\text{CPTG},e} R_e^2 / G$, retaining the common R_e dependence within the derived quantity. This is compared with

$$\log_{10}(M_{\text{dyn},e}/M_{\odot}) = 10.9356 \pm 0.0700. \quad (52)$$

Thus

$$\Delta \log_{10} M_e = +0.0048 \text{ dex}, \quad \Delta_{M,e}^{\text{nominal}} = +0.07\sigma. \quad (53)$$

Closure coordinate	Difference	Nominal independent-error residual
$g_{\text{CPTG},e} - g_{\text{dyn},e}$	$+1.634 \times 10^{-10} \text{ m s}^{-2}$	$+0.06\sigma$
$\log_{10}(M_{\text{CPTG},e}/M_{\odot}) - \log_{10}(M_{\text{dyn},e}/M_{\odot})$	$+0.0048 \text{ dex}$	$+0.07\sigma$

Table 3: One-radius CPTG closure against the correspondingly converted published virial dynamical comparator. The residuals are descriptive only because the support estimator and virial comparator share measured inputs, especially $\sigma_{\star}$ and R_e .

The closure is therefore numerically close, but its statistical interpretation remains limited by the shared R_e and $\sigma_{\star}$ inputs. The next section asks a different question that does not use the published morphology as an input: how is the solved radial field organized?

6 Structural Mode N and the Organization Length λ

The effective-radius closure asks whether the amplitude of the CPTG field is consistent with the available dynamical comparator. Structural Mode asks a different question: *where, across the galaxy, is the solved field changing most strongly?* The answer is summarized first by a physical organization length λ and only then by the dimensionless Structural Mode N .

The construction follows the same logic used in the dedicated CPTG Structural Mode study [9]:

$$\text{solved } g(r) \longrightarrow C_g(r) \longrightarrow w(r) \longrightarrow \lambda \longrightarrow N \longrightarrow \text{CSML}. \quad (54)$$

6.1 What λ measures

For a solved radial field,

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad \kappa_{\text{struct}} = 1 \text{ kpc}. \quad (55)$$

The quantity $C_g(r)$ combines the local field amplitude with the radial gradient of that field. Structural Mode does not use C_g itself as a morphology classifier. Instead it asks where C_g changes. The structural weight is

$$w(r) = \left| \frac{dC_g}{dr} \right|. \quad (56)$$

Regions in which the solved field structure changes rapidly receive more weight; regions in which it changes little receive less.

The organization length is the weighted radial centroid of that structural change over the registered outer interval:

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}. \quad (57)$$

Thus λ has a direct geometric meaning. It is the characteristic radius at which the registered radial structural change is centered. It is *not* a wavelength, a fitted scale length, a dark-halo parameter, or an additional degree of freedom. Once $g(r)$ and the registered interval are fixed, λ is determined.

Structural Mode removes the dimensional scale by comparing that centroid with the outer readout radius:

$$N = \frac{R}{\lambda}. \quad (58)$$

Because λ lies within $R/5 \leq \lambda \leq R$ whenever the structural weight is nonzero,

$$1 \leq N \leq 5. \quad (59)$$

The interpretation is immediate: a larger λ means that the structural change is weighted farther outward and therefore gives a smaller N ; a smaller λ means that the structural change is concentrated farther inward and therefore gives a larger N .

This is also why Structural Mode must not be confused with the cosmological phase:

$$N_\pi : \text{ when the galaxy is represented,} \quad N : \text{ where its solved field is organized.} \quad (60)$$

The downstream Curvature Structure Mode Index (CSMI) assigns a descriptive population label only after the continuous N is known. No observed Hubble type or visual morphology label is inserted into Eqs. (57)–(58).

For ZF-UDS-7329 the distinction becomes concrete. Because the ideal Sérsic reconstruction has no finite outer edge, this paper registers the outer Structural Mode readout radius at the projected semimajor-axis radius enclosing 95 percent of the model light. For a Sérsic profile with index n , the enclosed-light fraction is

$$\frac{L(< R)}{L_{\text{tot}}} = \frac{\gamma(2n, b_n(R/R_e)^{1/n})}{\Gamma(2n)}, \quad \frac{\gamma(2n, b_n)}{\Gamma(2n)} = \frac{1}{2}, \quad (61)$$

where b_n is the usual half-light normalization. The registered radius R_{95} is therefore determined by

$$\frac{\gamma(2n, b_n(R_{95}/R_e)^{1/n})}{\Gamma(2n)} = 0.95. \quad (62)$$

Using the adopted $n = 2.5$ and midpoint-converted $R_e = 0.90073$ kpc gives

$$R = R_{95} = 4.84240 \text{ kpc}. \quad (63)$$

The spherical-equivalent solution below gives

$$\lambda_{\text{sph}} = 1.45404 \text{ kpc} = 0.30027 R, \quad N_{\text{sph}} = 3.33030. \quad (64)$$

When the measured flattening is inserted into the baryonic source geometry, the edge-on oblate solution gives

$$\lambda_{\text{obl}} = 1.39593 \text{ kpc} = 0.28827 R, \quad N_{\text{obl}} = 3.46894. \quad (65)$$

The geometry correction therefore shifts the structural centroid inward by about 0.058 kpc, or roughly 58 pc, while leaving the registered outer radius unchanged. The increase in N is not an arbitrary classification change; it is the dimensionless expression of that inward shift in the solved field's radial organization.

For the numerical reconstruction, the radial source uses the midpoint-converted total stellar mass $\log_{10}(M_{\star}/M_{\odot}) = 11.1812 \pm 0.02$, the measured F277W Sérsic index $n = 2.5 \pm 0.1$, and a spatially constant stellar mass-to-light ratio. Every geometry branch below uses the same observed-epoch $a_{\star}$ from Eq. (38). No rotation velocity is inferred or fitted.

6.2 Spherical-equivalent baseline

The first calculation deliberately ignores the observed flattening. It Abel-deprojects the measured Sérsic profile as a spherical-equivalent source, providing a controlled baseline with

$$g_{\text{bar,sph}}(r) = \frac{GM_{\star}(< r)}{r^2}. \quad (66)$$

The same CPTG response law used in the effective-radius closure is then solved over radius. Because an ideal Sérsic profile has no finite edge, the registered structural readout retains the projected 95-percent-light radius defined above. For the central reconstruction, $C_g(r)$ is monotonically decreasing throughout $R/5 \leq r \leq R$. Hence $w(r) = -dC_g/dr$ on the registered interval, and the continuous centroid is evaluated with the endpoint identities

$$\begin{aligned} \int_{R/5}^R w(r) dr &= C_g(R/5) - C_g(R), \\ \int_{R/5}^R r w(r) dr &= \frac{R}{5} C_g(R/5) - R C_g(R) + \int_{R/5}^R C_g(r) dr. \end{aligned} \tag{67}$$

This endpoint-controlled form is algebraically identical to the continuous definition on the monotone interval and avoids dependence on whether a numerical grid lands exactly on the integration boundaries.

The spherical-equivalent central result is

$$\lambda_{\text{sph}} = 1.45404 \text{ kpc}, \quad N_{\text{sph}} = 3.33030. \tag{68}$$

Under the current descriptive CSMI intervals, $3.30 < N \leq 3.55$ maps to Bulged Spiral. The central value lies 0.03030 above the Early-Spiral/Bulged-Spiral boundary. None of the one-at-a-time quoted source perturbations crosses $N = 3.30$ in the spherical surrogate, although the high-dispersion and low-Sérsic-index cases remain closest to that boundary.

Structural case	N	CSMI readout
Central: $\sigma_{\star} = 250 \text{ km s}^{-1}$, $n = 2.5$	3.33030	Bulged Spiral
$\sigma_{\star} = 230 \text{ km s}^{-1}$	3.35654	Bulged Spiral
$\sigma_{\star} = 270 \text{ km s}^{-1}$	3.30378	Bulged Spiral
$n = 2.4$	3.30965	Bulged Spiral
$n = 2.6$	3.34667	Bulged Spiral
$\log_{10} M_{\star} - 0.02 \text{ dex}$	3.32259	Bulged Spiral
$\log_{10} M_{\star} + 0.02 \text{ dex}$	3.33781	Bulged Spiral
$R_e = 0.89083 \text{ kpc}$	3.33581	Bulged Spiral
$R_e = 0.91063 \text{ kpc}$	3.32483	Bulged Spiral

Table 4: Observed-epoch Structural Mode sensitivity under the spherical-equivalent Sérsic reconstruction. The numerical result is stable and the listed one-at-a-time source perturbations remain above the current $N = 3.30$ CSMI boundary.

Grid convergence from 4000 to 32000 radial support points changes the spherical central result by less than 10^{-7} in N , so this boundary sensitivity is not a radial-grid artifact.

6.3 Measured flattening: oblate-source geometry stress test

The spherical baseline does not use all of the published source geometry. Carnall et al. fitted the F277W image with PetroFit in combination with Astropy [2, 3, 4]. In the

Astropy Sérsic parameterization used by PetroFit, ellipticity is

$$e = 1 - \frac{b}{a}, \quad (69)$$

so the reported $e = 0.77 \pm 0.01$ corresponds to the projected axis ratio

$$q_{\text{proj}} \equiv \frac{b}{a} = 0.23 \pm 0.01. \quad (70)$$

The fitted Sérsic radius is the elliptical-radius parameter; along the projected major axis its half-light intercept is the reported R_e . An independent multi-band analysis likewise finds a strongly flattened, nearly edge-on disc-like light distribution, with F444W $b/a = 0.29 \pm 0.01$ [5]. This morphology is used only as an external geometry cross-check and is not substituted for the F277W source inputs above.

The cleanest geometry stress test changes only the source shape. Without introducing a rotation curve, Jeans fit, or additional free normalization, consider a family of similar oblate stellar-density surfaces

$$m^2 = R^2 + \frac{z^2}{q_{\text{int}}^2}, \quad 0 < q_{\text{int}} \leq 1, \quad (71)$$

with constant intrinsic axis ratio q_{int} and the same observed semimajor-axis Sérsic profile. For an oblate system observed at inclination i [6],

$$q_{\text{proj}}^2 = \cos^2 i + q_{\text{int}}^2 \sin^2 i. \quad (72)$$

Thus the measured $q_{\text{proj}} = 0.23$ admits $76.7029^\circ \lesssim i \leq 90^\circ$; the edge-on endpoint has $q_{\text{int}} = q_{\text{proj}} = 0.23$, while $q_{\text{int}} \rightarrow 0$ as i approaches the lower geometric limit.

The deprojection can be normalized without an additional shape parameter. Let $S(s)$ denote the measured semimajor-axis Sérsic shape and define

$$L_S = 2\pi \int_0^\infty S(s)s \, ds, \quad \Sigma_\star(s) = \frac{M_\star}{q_{\text{proj}}L_S} S(s). \quad (73)$$

For similar oblate spheroids, the corresponding constant- M/L density is [6]

$$\rho_q(m) = -\frac{M_\star}{\pi q_{\text{int}}L_S} \int_m^\infty \frac{dS/ds}{\sqrt{s^2 - m^2}} ds. \quad (74)$$

The factor q_{int}^{-1} fixes the same total stellar mass for every allowed intrinsic thickness; it is not a fitted density normalization. The exact equatorial source field of nested similar oblate homoeoids is then [6]

$$g_{\text{bar,obl}}(R) = \frac{4\pi G q_{\text{int}}}{R} \int_0^R \frac{\rho_q(m)m^2 \, dm}{\sqrt{R^2 - (1 - q_{\text{int}}^2)m^2}}. \quad (75)$$

For $q_{\text{int}} = 1$, Eq. (75) reduces algebraically to the spherical $GM(< R)/R^2$ source. The CPTG response itself is not altered: Eq. (75) replaces only the baryonic source field

entering the same observed-epoch radial response law, with $a_\star$ still supplied by the declared pressure-support estimator. This is therefore an axisymmetric *source-geometry stress test*; it is not a Jeans model, a fitted rotation solution, or a new two-dimensional definition of Structural Mode.

For the edge-on endpoint $q_{\text{int}} = 0.23$, the solved structural coordinate becomes

$$\lambda_{\text{obl}} = 1.39593 \text{ kpc}, \quad N_{\text{obl}} = 3.46894. \quad (76)$$

The measured flattening therefore moves the mode upward by

$$\Delta N = N_{\text{obl}} - N_{\text{sph}} = 0.13864, \quad (77)$$

rather than pushing the solution downward through the $N = 3.30$ boundary.

Geometry case	q_{int}	N	CSMI readout
Spherical-equivalent baseline	1.00000	3.33030	Bulged Spiral
Oblate edge-on central	0.23000	3.46894	Bulged Spiral
Oblate $i = 80^\circ$ central	0.15315	3.48644	Bulged Spiral
Oblate $i = 77^\circ$ central	0.04919	3.51164	Bulged Spiral
Razor-thin numerical limit	$\rightarrow 0$	3.52425	Bulged Spiral

Table 5: Non-fitted oblate-source geometry stress test at the central measured source values. The projected axis ratio fixes a family of intrinsic thickness/inclination pairs rather than a unique deprojection. Across that central family the current CSMI lookup remains inside the Bulged-Spiral interval.

At the edge-on endpoint, one-at-a-time quoted uncertainties give $N = 3.44701\text{--}3.49006$ for $\sigma_\star$, $3.45103\text{--}3.48282$ for n , $3.46263\text{--}3.47505$ for total stellar mass, $3.46409\text{--}3.47382$ for R_e , and $3.46673\text{--}3.47117$ for the measured ellipticity. A deterministic simultaneous signed-corner exercise, taking all quoted source quantities in the directions that minimize or maximize N while retaining the edge-on identification $q_{\text{int}} = q_{\text{proj}}$, gives

$$3.41582 \leq N_{\text{edge-on,corner}} \leq 3.51724. \quad (78)$$

This corner range is a sensitivity envelope, not a confidence interval: the source measurements are not asserted to be statistically independent and no covariance model is supplied.

The central oblate inclination family itself remains below the next CSMI boundary, approaching $N \simeq 3.52425$ as $q_{\text{int}} \rightarrow 0$. If observational extremes, unknown intrinsic thickness, and inclination are combined simultaneously, the deliberately compound high- N quoted-input corner reaches $N \simeq 3.56642$ in the $q_{\text{int}} \rightarrow 0$ numerical limit, only slightly above the current $N = 3.55$ boundary. This is a stress point, not a confidence bound or preferred deprojection, and no probability should be assigned to it without a joint geometry and kinematic posterior. Continuous N is therefore the primary result. The geometry audit establishes the narrower but important conclusion that the spherical value is conservative with respect to the $N = 3.30$ boundary: inserting the measured flattening into the baryonic source field increases the solved mode.

The oblate calculation is numerically stable. At fixed 480-point Gauss–Legendre source quadrature, changing the radial support grid from 4000 to 32000 points changes the

edge-on result by less than 2×10^{-11} in N . Varying the source quadrature from 320 to 800 points changes N by less than 3×10^{-9} . These numerical effects are negligible compared with the source and geometry systematics.

The geometry calculation therefore answers the immediate concern raised by the strongly elongated image: the spherical solution does not obtain its high mode by artificially concentrating the source. Representing the measured flattening moves the solved mode farther upward in N .

7 Results, Interpretation, and Limitations

The paper answers three different questions that should not be conflated. The native branch specifies the cosmological phase and shows that geometric phase accumulates much more rapidly per gigayear in the early universe. The effective-radius solve tests the amplitude of the bound-system field at the observed phase without invoking a dark halo. Structural Mode then uses λ to identify where that solved field is organized radially. The clock, local field amplitude, and radial organization are separate CPTG coordinates evaluated for the same galaxy.

The calculation has three principal results. First, the EXCELS redshift places ZF-UDS-7329 at $N_\pi = -1.433988687$ on the common native phase, while the central formation marker gives $N_{\pi,\text{form}} = -2.74084$ and a formation-to-observation native phase cadence 6.97 times the observation-to-present cadence. Second, the one-radius support measurement gives $a_\star \simeq 2.2487 \times 10^{-9} \text{ m s}^{-2}$ and a CPTG effective-radius field nominally $+0.06\sigma$ from the correspondingly converted published virial comparator. Third, the Structural Mode calculation gives $N_{\text{sph}} = 3.33030$ for the spherical baseline and $N_{\text{obl}} = 3.46894$ for the measured-flattening edge-on case. These results answer different questions and should not be collapsed into one significance statement.

The effective-radius agreement is a closure result. The published virial mass uses the same compact radius and stellar velocity dispersion that enter the CPTG support estimator. The quoted $+0.06\sigma$ field residual and $+0.07\sigma$ mass residual are therefore descriptive closure metrics under an independent-error approximation, not frequentist significances of an independent test.

The Structural Mode result has different limitations. The spherical-equivalent baseline is numerically stable at $N = 3.33030$, and the listed one-at-a-time source perturbations remain above the $N = 3.30$ boundary. Carrying the measured projected flattening into an oblate source raises the edge-on central result to $N = 3.46894$, and the central inclination/thickness family remains below the next $N = 3.55$ boundary while approaching $N \simeq 3.52425$ in the razor-thin limit. The intentionally compounded high- N stress corner can reach $N \simeq 3.56642$, so the hard CSMI label should still not be treated as a universal uncertainty statement. Continuous N is the primary result; the robust geometry statement is directional: representing the measured flattening increases N relative to the spherical surrogate.

A second Structural Mode systematic is the declared spatially constant stellar mass-to-light ratio. The EXCELS source paper constrains the integrated stellar population,

while the independent multi-band morphology analysis indicates radial population/color structure [1, 5]. The exact continuous N reported here is therefore conditional on the present light-traces-mass reduction. A future resolved stellar-mass map can replace that assumption directly in the baryonic source without changing the definition of λ or N .

What remains missing dynamically is resolved kinematics. Imaging alone does not determine a unique intrinsic thickness or inclination, and the integrated σ_* cannot separate random pressure support from ordered rotation. A stronger future test should therefore combine the measured surface-brightness or stellar-mass geometry with spatially resolved stellar kinematics, introducing a rotation-aware support coordinate only if rotation is actually measured. That would convert the present source-geometry stress test into a genuinely axisymmetric dynamical test.

Finally, the cosmological branch clocks have only a placement and native-history role in the bound-system calculation. They select and describe the phase and its cadence; they do not define an $a_*(t)$ law or rescale the observed object-dependent a_* merely because the galaxy is observed at high redshift.

8 Conclusion

ZF-UDS-7329 illustrates why the conventional phrase “too massive, too early” has a different meaning in CPTG. The EXCELS source paper gives a central inferred formation marker $z_{\text{form}} = 14.5^{+8.9}_{-4.2}$, quenching marker $z_{\text{quench}} = 9.90^{+7.79}_{-3.29}$, and stellar mass $\log_{10}(M_*/M_\odot) = 11.19 \pm 0.02$ [1]. On the CPTG native branch that formation marker lies at $N_{\pi, \text{form}} = -2.74084$ and $t_{\pi, \text{form}} = 0.21451$ Gyr. The average native phase cadence from formation to observation is about 6.97 times the observation-to-present cadence, placing the early assembly deep in the high-cadence native regime.

At the observed phase, $z = 3.1954 \pm 0.0002$ gives $N_\pi = -1.433988687$. The published EXCELS intrinsic stellar dispersion and F277W geometry give the one-radius estimator

$$a_* \simeq a_\sigma = (2.2487 \pm 0.3606) \times 10^{-9} \text{ m s}^{-2}. \quad (79)$$

Using the stellar source, the reduced effective-radius solve gives

$$g_{\text{CPTG}, e} = (1.4975 \pm 0.0741) \times 10^{-8} \text{ m s}^{-2}, \quad (80)$$

compared with

$$g_{\text{dyn}, e} = (1.4811 \pm 0.2420) \times 10^{-8} \text{ m s}^{-2}. \quad (81)$$

The nominal $+0.06\sigma$ field residual is a consistency closure only because the two constructions share measured inputs.

The Structural Mode calculation gives

$$N_{\text{sph}} = 3.33030, \quad N_{\text{obl}, \text{edge}} = 3.46894. \quad (82)$$

The spherical-equivalent result remains above the $N = 3.30$ boundary under the listed one-at-a-time source perturbations. For the central measured inputs, every allowed oblate

inclination/thickness case moves the mode upward, approaching $N \simeq 3.52425$ in the razor-thin limit. The current CSMI label remains a downstream descriptive lookup; continuous N carries the scientific result.

The remaining limitations are physical rather than numerical. The exact radial source presently assumes spatially constant stellar M/L , and a unique axisymmetric dynamical solution requires resolved kinematics to determine inclination, intrinsic thickness, and any ordered rotation. Neither limitation alters the present conclusions: the EXCELS source is consistently placed on the native phase, the one-radius CPTG field closes against the published virial comparator within the stated shared-input limits, and the high Structural Mode survives explicit treatment of the galaxy’s strong observed flattening.

Data, Code, and Evidence Availability

The population and phase values are drawn from the EXCELS source paper for PRIMER-EXCELS-55410/ZF-UDS-7329 [1]. The authors provide machine-readable results and supporting material at https://github.com/HinLeung622/EXCELS_massive_quiescent. The imaging and kinematic quantities used in the bound-system calculation are the published EXCELS measurements for the same galaxy [2]. The independent multi-band morphology study is used as an external geometry and radial-population cross-check and does not replace the adopted F277W structural values [5].

The accompanying CPTG evidence package is

`CPTG_ZF-UDS-7329_Evidence.zip`

It reproduces the native phase and branch clocks, source-coordinate transformations, stellar-source construction, pressure-support estimator, local curvature-transport radius, implicit effective-radius field solve, nominal closure residuals, Sérsic R_{95} registration, spherical deprojection, oblate-homoeoid source reconstruction, endpoint-controlled Structural Mode evaluation, inclination/thickness scan, quoted-input sensitivity envelopes, and numerical-convergence tests. Numerical outputs retain guard digits for reproducibility; manuscript values are rounded to the precision warranted by the observations.

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